



University of Puerto Rico

Mayaguez campus

Department of Computer Science and Engineering

Undergraduate research in software (INSO 4998 - 001)

Second Report

Task 3: Attempting to take mid-circuit measurements

Juan D. Guadalupe Rosado

Daniel C. Colon Rivera

Professor Venkataramani Kumar

For this task, we were instructed to attempt to take the measurements regarding the outputs of all the gates within a constructed circuit. After attempting to do so, I was met with the error “StateVectorSampler cannot handle mid-circuit measurements” indicating that said operations are not possible. Studying the problem, I reached the conclusion that this result is to be expected, as any attempt to measure the state of the system, at any point of the circuit, will immediately collapse the state vector into one of the outputs, which breaks the circuit. I attempted to research ways of probing the circuit for results without provoking the previously mentioned error, which lead me to a promising term called weak measurement. Weak measurements is a way to gently probe for the state of my qubits without causing it to collapse into a given value. In other words, by extracting minimal information from my gates over many runs, I can begin to build a distribution on what values the qubits could take when they are affected by a gate. This would entail having a “coupling strength” small enough that the result is not significant on its own, which would require many runs to correctly profile a state. Sadly, Qiskit does not have operations to carry this out, which means that we would have to build our own solution. Admittedly, I am not well versed in the topic of weak measurement, I would need to further research it to see if my understanding is correct, however, from what I have grasped, weakly measuring a circuit might lead to what we want, which is mid-circuit readings. What I propose is that we build distributions for a gate that is going to be used, then we take the probabilities generated by that distribution and use it as the state vector that will serve as input for the next gate. We repeat this until we have no more gates to run through, thus, theoretically, giving us a window into the outputs of each gate mid circuit. A problem that arises with this proposed method is that we would need to iterate over each separate gate to build the “distribution profile” needed, which means that we need to run each gate for the number of iterations we deem fit, costing a lot computing resources to carry out, as we would need to wait until all operations are done. I will be bringing this matter in our next meeting, as I see potential in the idea and would love to hear what the others think.